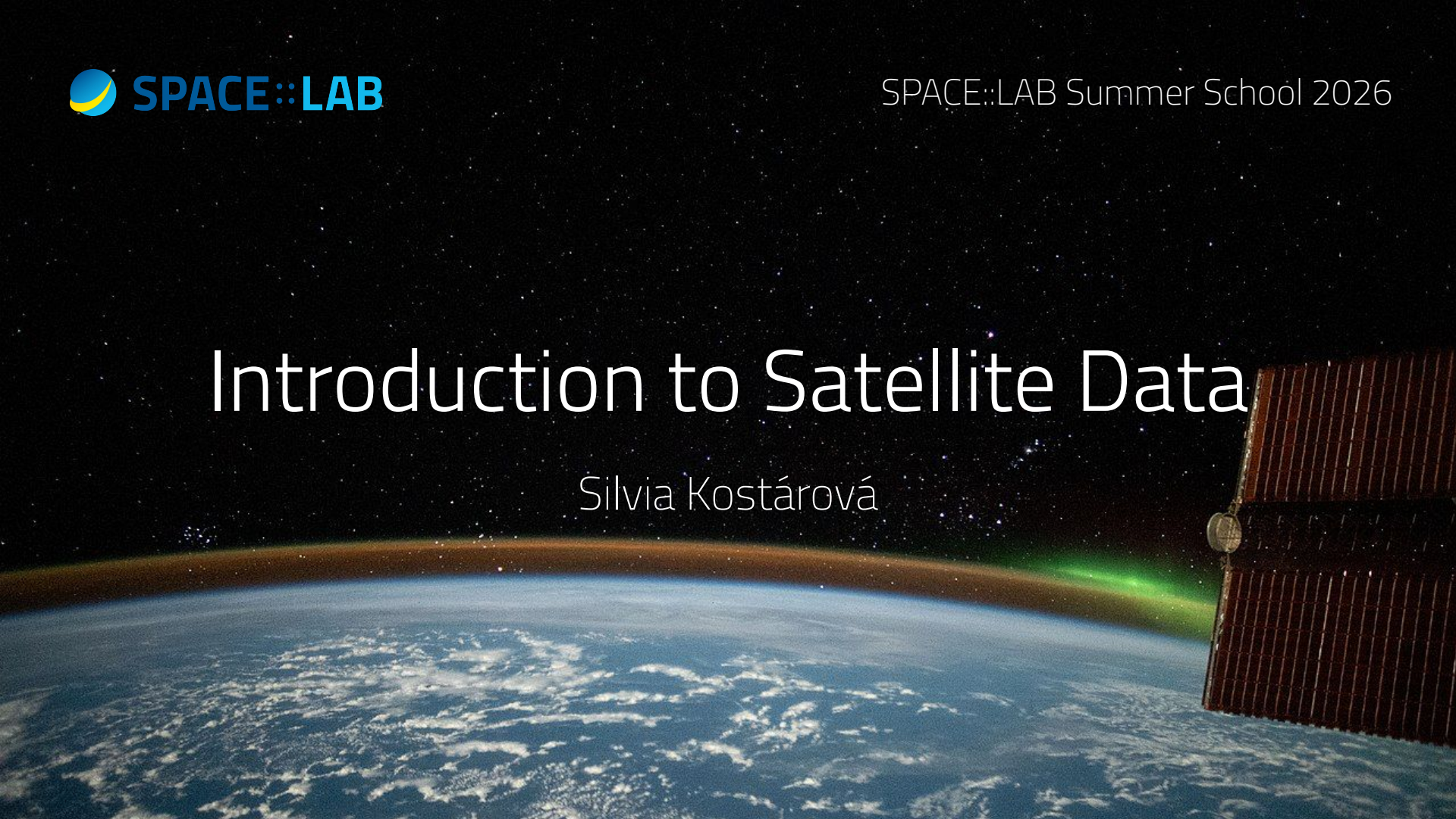
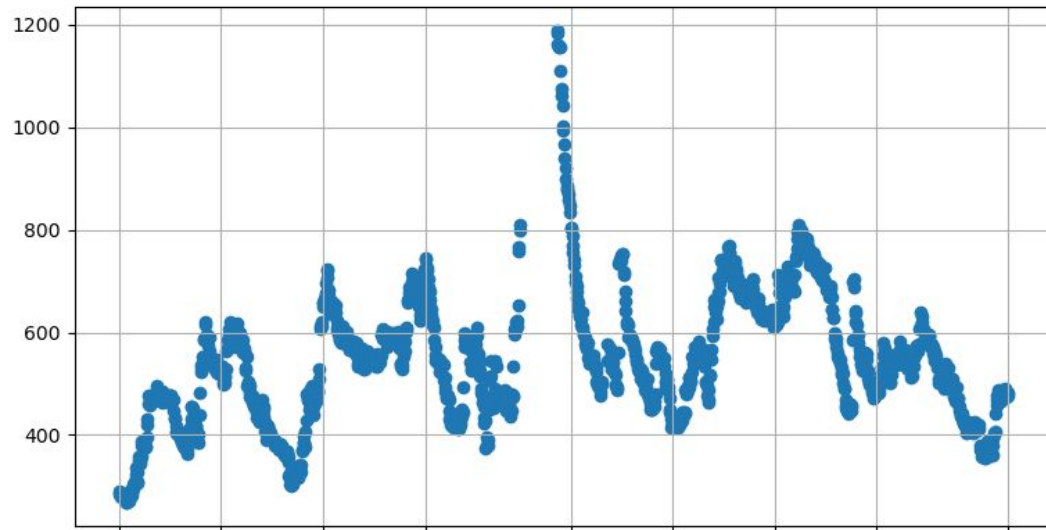


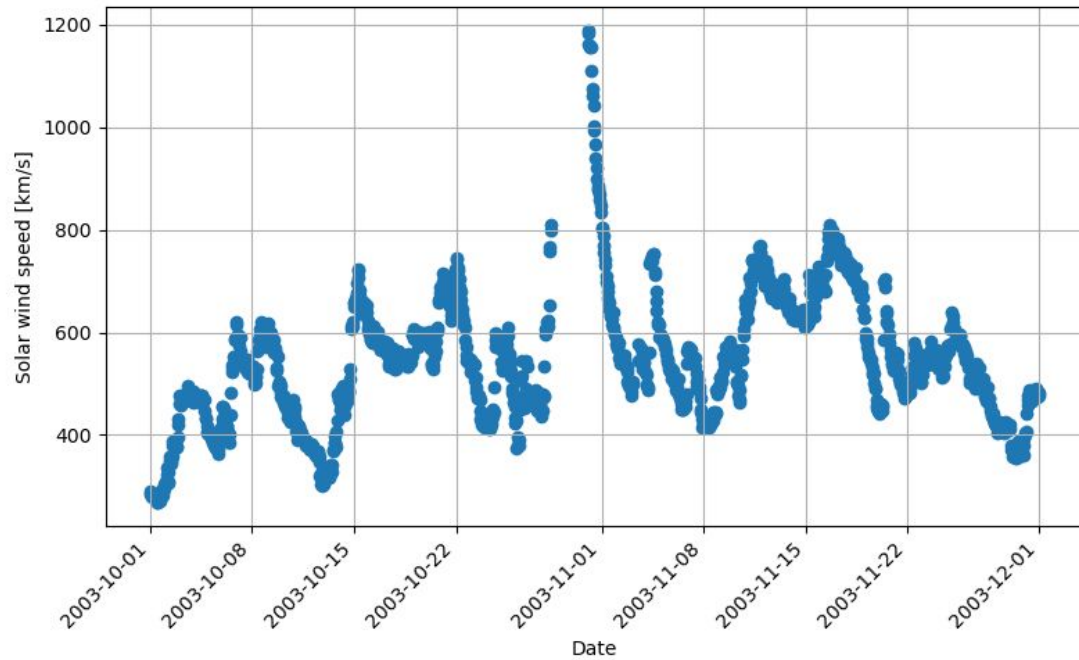
Introduction to Satellite Data

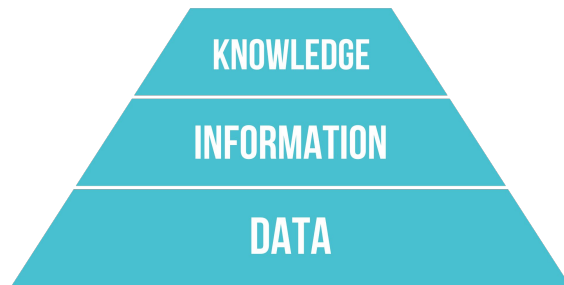
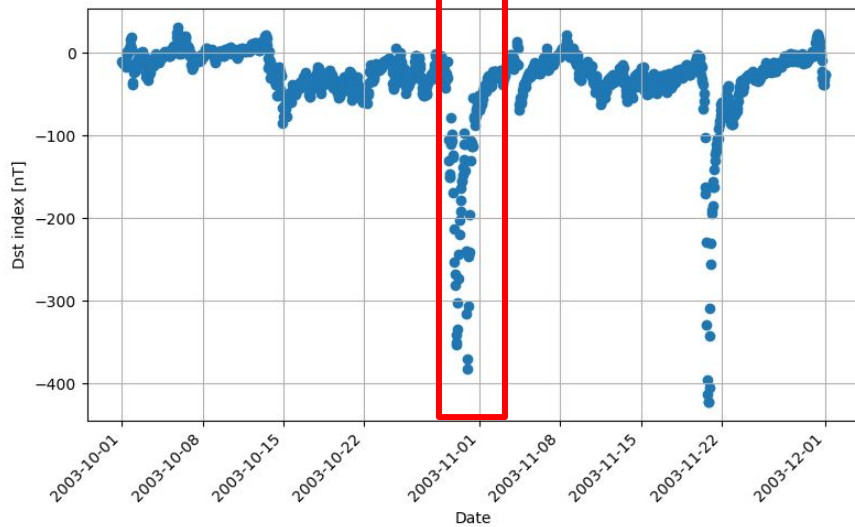
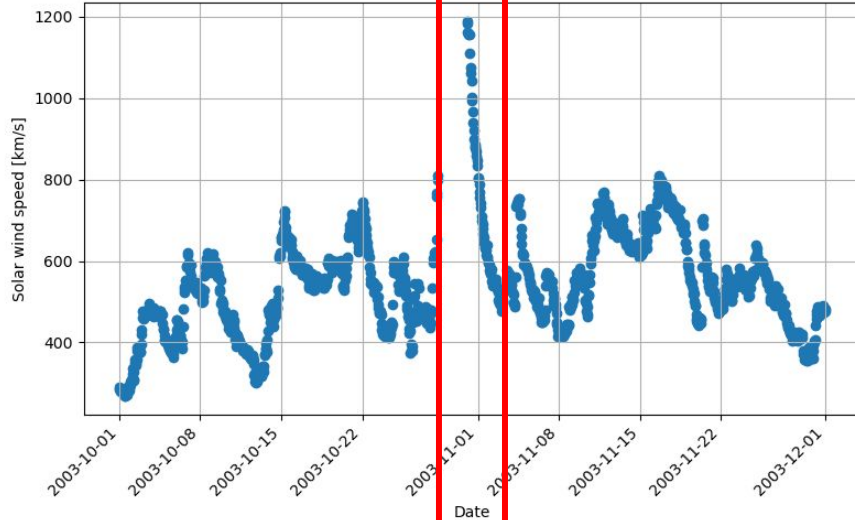
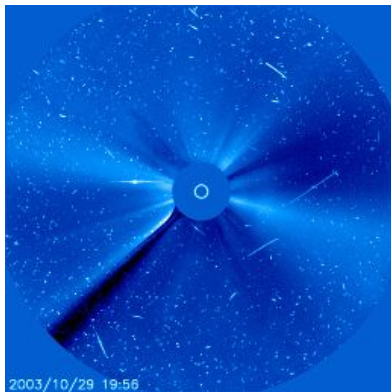
Silvia Kostárová

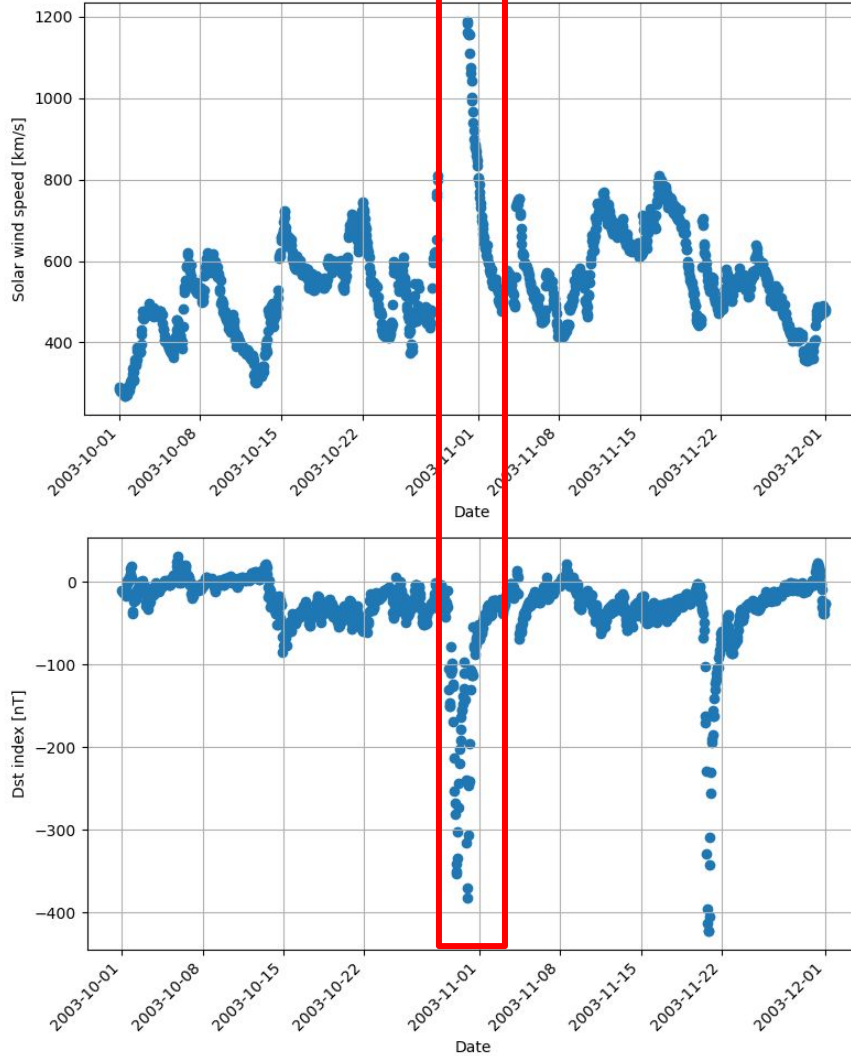
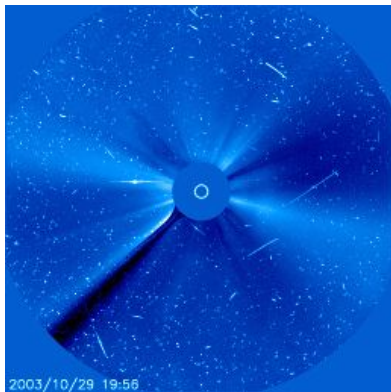




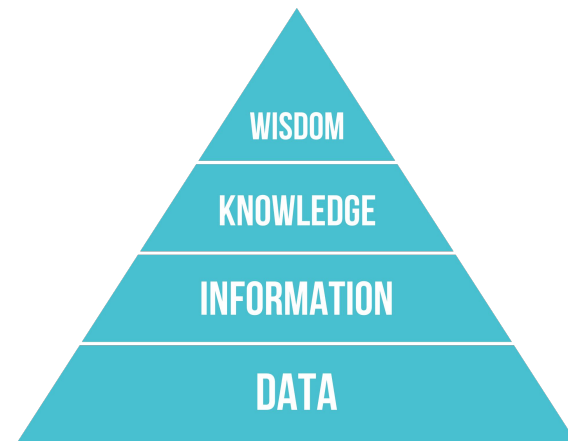
DATA



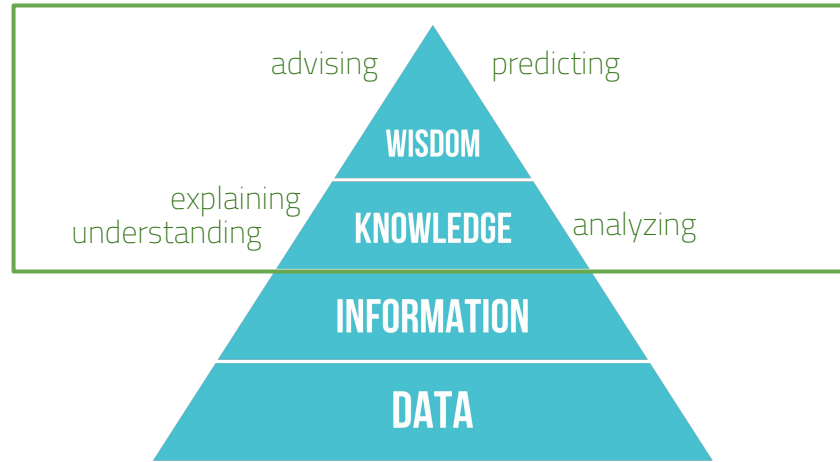




→ Shut down sensitive instruments on satellites

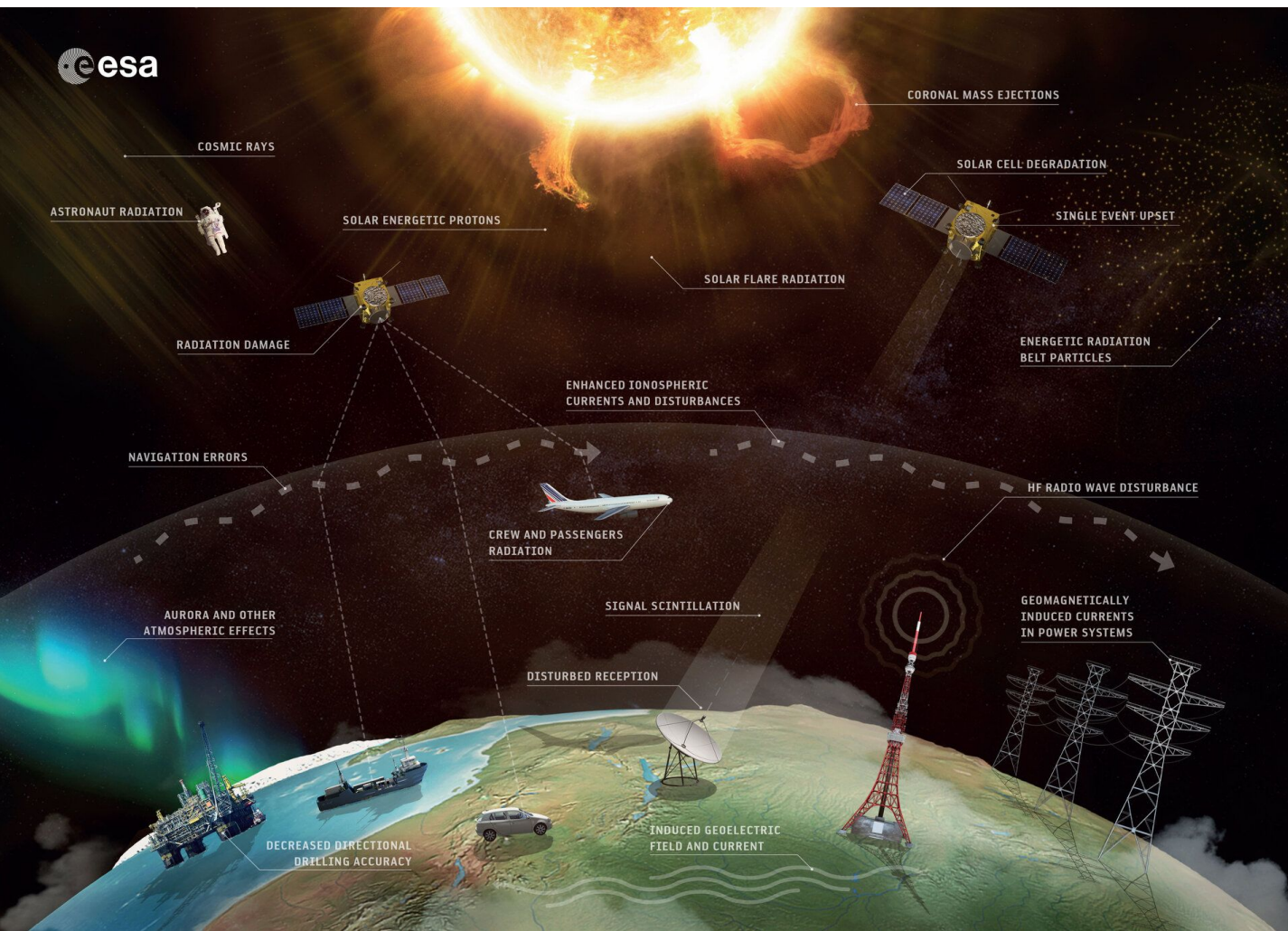


As a scientist...



... I need

↑
satellite observations

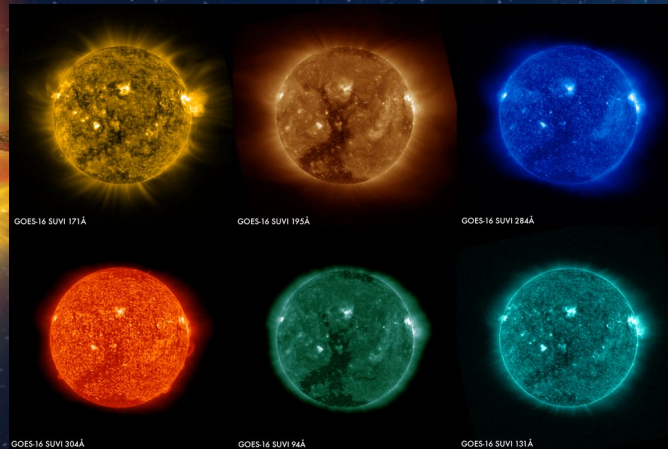


heliophysics monitoring instruments

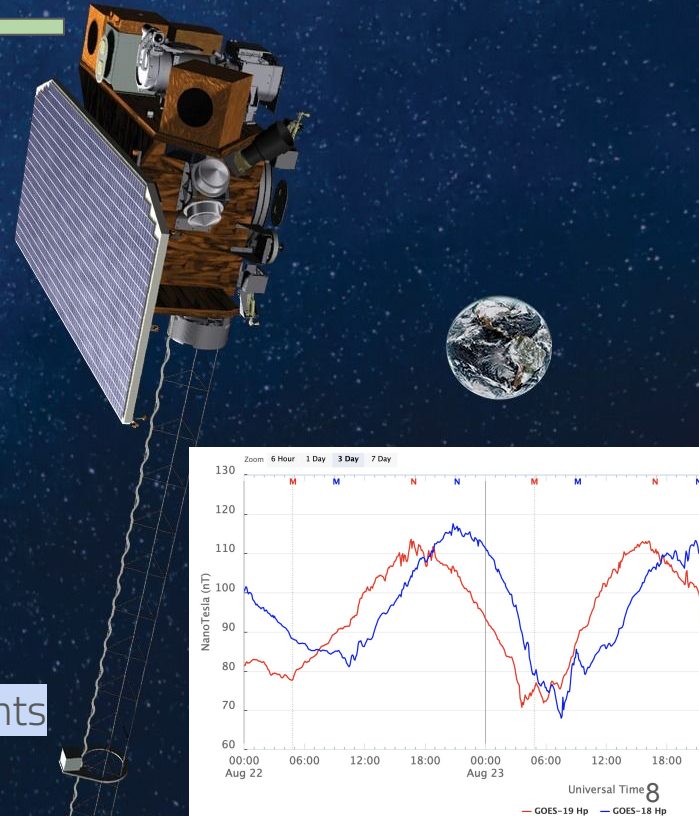
SWARM (ICON, GOLD)

airglow cameras,
Doppler sounding system, GNSS receivers

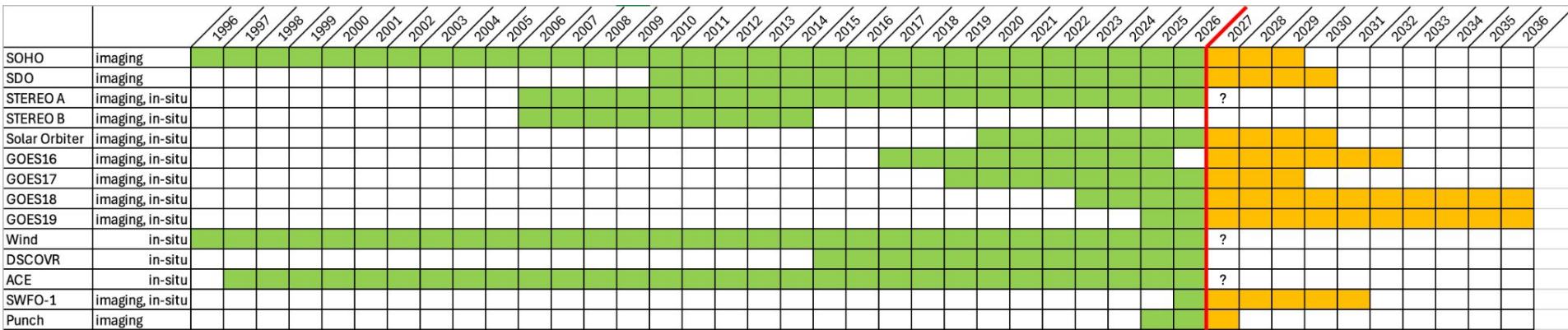
Remote sensing



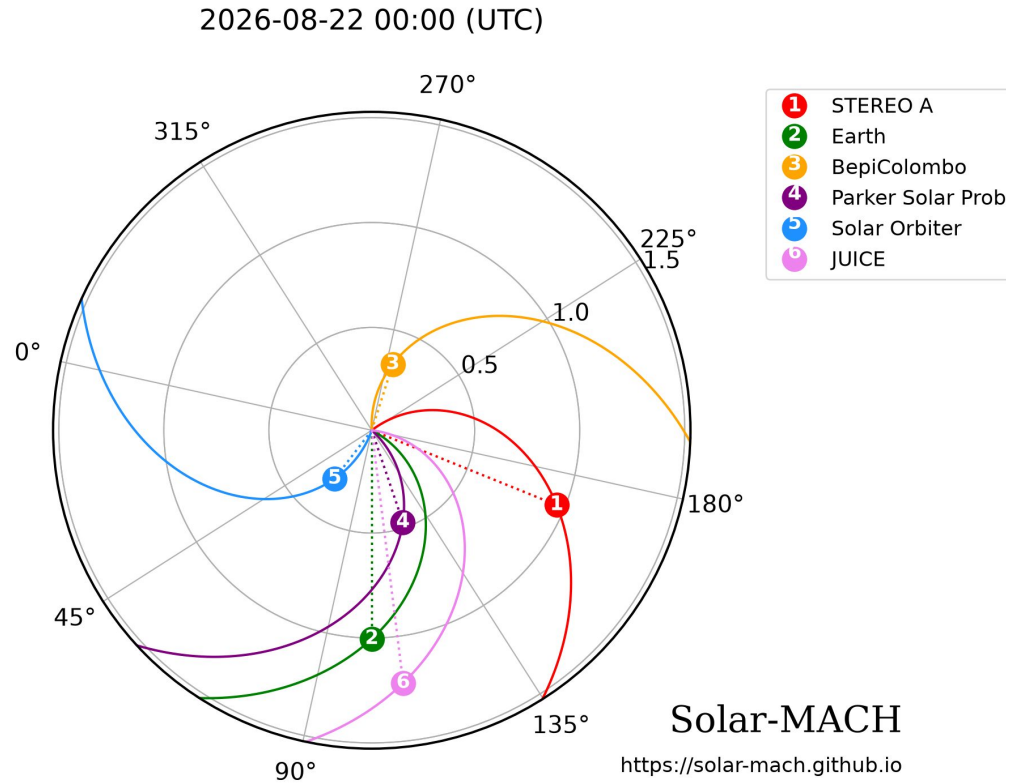
In situ measurements



Missions Timeline



Where are they now



Data Levels

Level 0 (L0)

- raw telemetry, raw binary packets
- it consists of uncalibrated "instrument counts" (detector pulses or raw voltages)

Level 1 (L1)

- raw data converted into standard engineering and physical units
- precise orbital coordinates

Level 2 (L2)

- fully processed, science-ready data products stripped of instrumental bias, representing the actual state of the space environment.

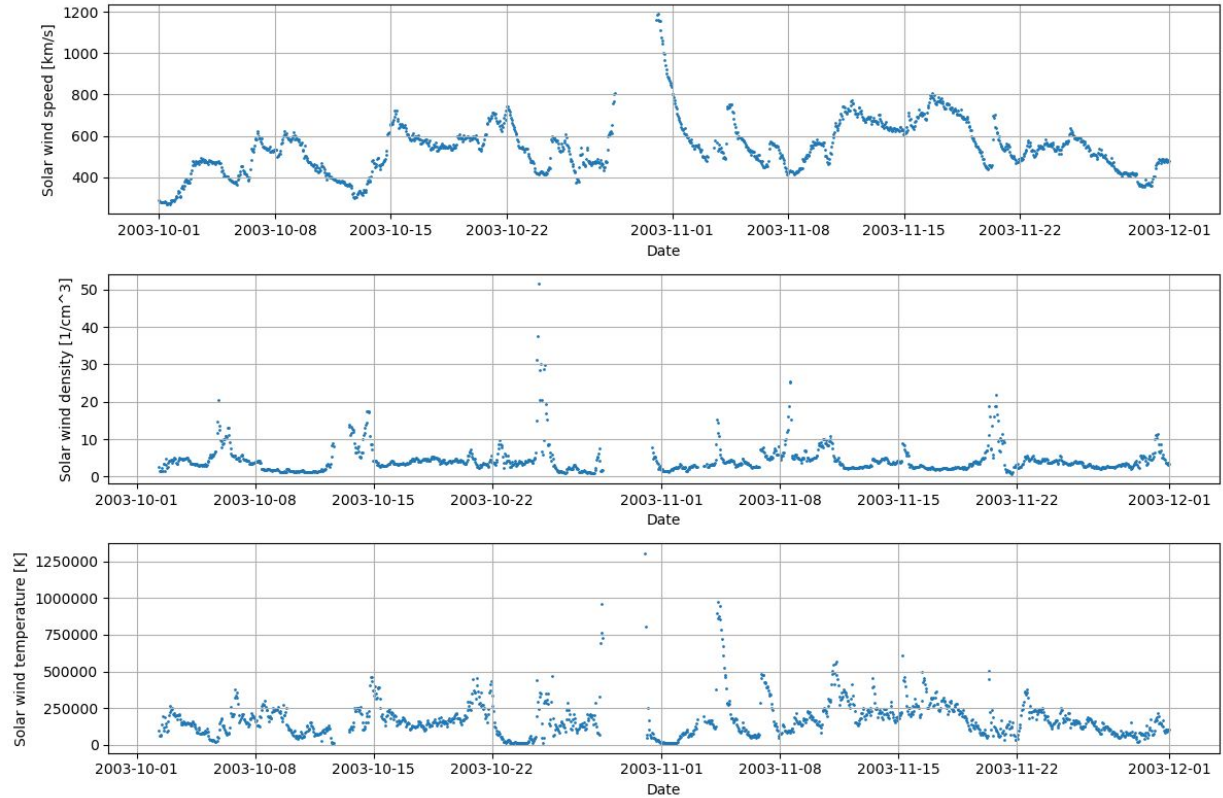
Solar wind data

Solar wind plasma

velocity/speed v_{sw} [km/s]

density n_{sw} [particles/cm³]

temperature T_{sw} [K]



Solar wind data

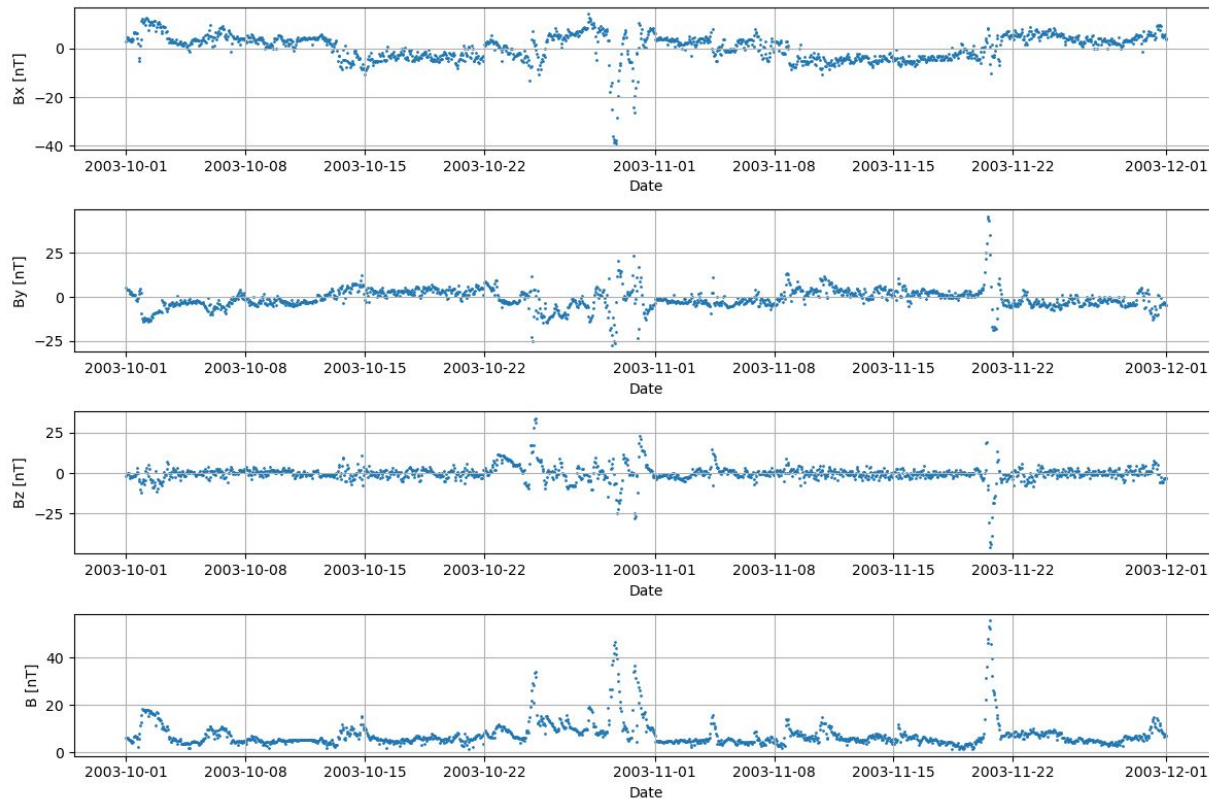
Interplanetary magnetic field (IMF)

B_x [nT]

B_y [nT]

B_z [nT]

B_{total} [nT]



How to access the data

1. OMNIWeb <https://omniweb.gsfc.nasa.gov/form/dx1.html>
 - Wind & ACE
2. Searching for the official mission and instrument page
3. HAPI (Heliophysics Data Application Programmer's Interface)
<https://hapi-server.org/servers/>

Data preprocessing

depends on your goal

- **Data Cleaning:** Removes duplicates, handles missing values, manages extreme outliers.
- **Data Integration:** Combines data from multiple sources
- **Data Construction:** Generates derived variables
- **Data Transformation:** Converts values into suitable formats via normalization, standardization, or categorical encoding
- **Data Splitting:** Divides data into training, validation, and test sets